Part A

Practical Exercise 2: Circuit Design

Part A

1. Circuit (a)

voltage = current × resistance

resistance = 20+30+50
= 100Ω

Circuit (b)

voltage = current × resistance

resistance =
$$\left(\frac{1}{20}+\frac{1}{100}+\frac{1}{50}\right)^{-1}$$
$$=\left(\frac{5+1+2}{100}\right)^{-1}$$
$$=\left(\frac{8}{100}\right)^{-1}$$
= 100/8
= 12.5Ω

2. circuit (a)

voltage = current × resistance

$$\text{current} = \frac{125}{100}$$
= 1.25A

circuit (b)

voltage = current × resistance

$$\text{current} = \frac{125}{12.5}$$
= 10A

3. circuit (a)

- resistors in series
- current is equal

current in all resistors = 1.25A

circuit (b)

- resistors in parallel
- voltage is equal

voltage = current × resistance

At 20Ω
$$\text{current} = \frac{125}{20}$$
= 6.25A

At 100Ω
$$\text{current} = \frac{125}{100}$$
= 1.25A

At 50Ω
$$\text{current} = \frac{125}{50}$$
= 2.5A

4. circuit (a)

- resistors in series
- current is equal

voltage = current × resistance

At 20Ω
voltage = 1.25×20
= 25V

At 30Ω
voltage = 1.25×30
= 37.5V

At 50Ω
voltage = 1.25×50
= 62.5V

circuit (b)

- resistors in parallel.
- voltage is equal

voltage in each resistor = 125V

5. circuit (a)

power = voltage × current

At 20Ω
power = 25×1.25
= 31.25W

At 30Ω
power = 37.5×1.25
= 46.875W

circuit (b)

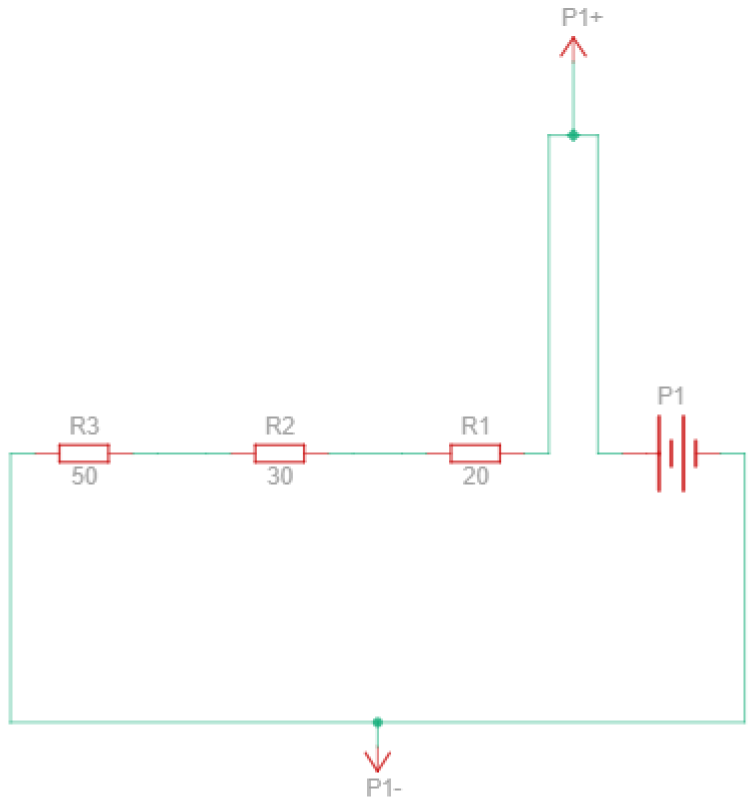
power = voltage × current

At 20Ω
power = 125×6.25
= 781.25W

At 100Ω
power = 125×1.25
= 156.25W

Part B

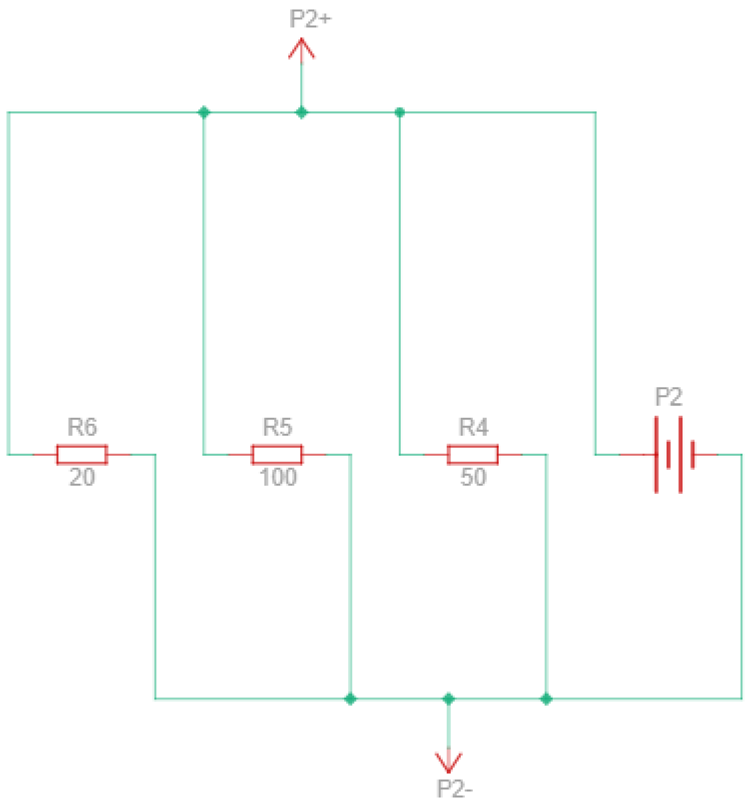
Circuit Diagram 1



Component List

1. Power Supply (P1): 125V, 1.25A
2. Resistor (R3): 50Ω
3. Resistor (R2): 30Ω
4. Resistor (R1): 20Ω
5. Jumper wires: 4

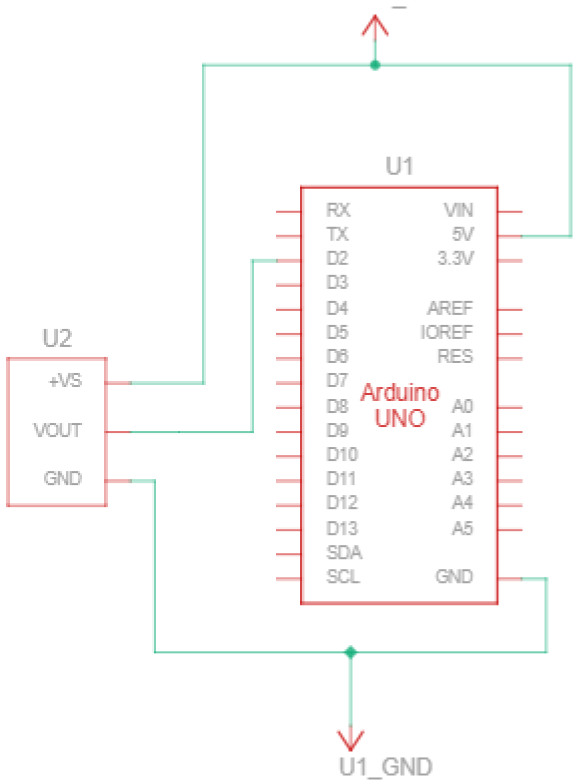
Circuit Diagram 2



Component List

1. Power Supply (P2): 125V, 12.5A
2. Resistor (R6): 20Ω
3. Resistor (R5): 100Ω
4. Resistor (R4): 50Ω
5. Jumper wires: 6

Circuit Diagram 3



Component List

1. Arduino Uno (used in place of Arduino Nano since Tinkercad doesn't have Arduino Nano)
2. Breadboard (small)
3. Temperature Sensor
4. Jumper wires: 5